

graph viewers), type:

```
sudo pip install graphviz
```

- The code for installing Keras is:

```
sudo pip install keras
```

The Keras config file can be found at `~/.keras/keras.json`, which has information like *backend: tensorflow*.

Installation in Windows 10

It is possible to install Keras in Windows 10 also, using Anaconda. The following steps need to be followed.

- Download Anaconda <https://www.anaconda.com/distribution/#download-section>.
- While installing Anaconda, you may optionally want to download the VSCode also as the preferred IDE for writing DL programs. However, you can use the Jupyter notebook as well, initially, for writing and running the program, piece-wise.
- Next launch Anaconda Prompt from the Windows program launcher.
- Next install Python-3.6:

```
conda install python=3.6
```

- Create a conda environment:

```
conda env create -q -f environment.yml
```

- A typical *environment.yml* looks like:

```
name: practice
channels:
- defaults
dependencies:
- python=3.7.*
- bz2file==0.98
- cython==0.29.*
- pip==19.1.*
- numpy==1.16.*
- jupyter==1.0.*
- matplotlib==3.1.*
- setuptools==41.0.*
```

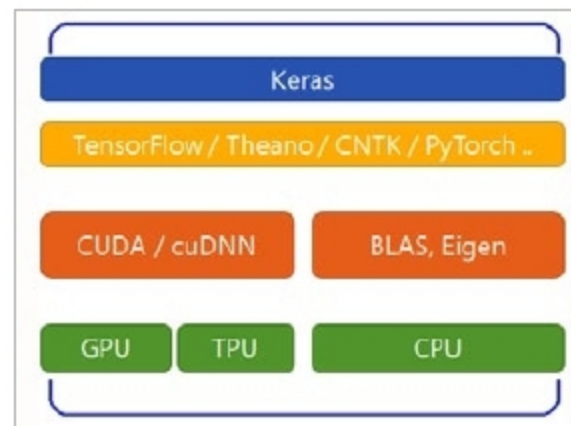


Figure 4: Layers of a DL deployment taxonomy

```
- scikit-learn==0.21.*
- scipy==1.2.*
- pandas==0.24.*
- pillow==6.1.*
- seaborn==0.9.*
- h5py==2.9.*
- pytest==5.0.*
- twisted==19.2.*
- tensorflow==1.13.* // For GPU
it will be something like tensorflow-gpu==1.13.*
- pip:
  - keras==2.2.*
```

- `conda info --envs` will show the practice as the environment.
- `conda activate practice`
- jupyter notebook
- Launch `http://localhost:8888/notebooks` in your favourite browser.

A sample program using Keras

We are now going to look at the MNIST data set digit-detection problem using Jupyter Notebook.

To get an idea about what is inside the MNIST data set, use the following code:

```
import keras
from keras.datasets import mnist
from keras import models, layers
import numpy as np
import matplotlib.pyplot as plt
from keras.utils import to_categorical

# Reading the mnist dataset which is
# 28X28 pixel of 70k integer (0-9) RGB
# scale images
# Out of these 70K, I am using 60K to
```

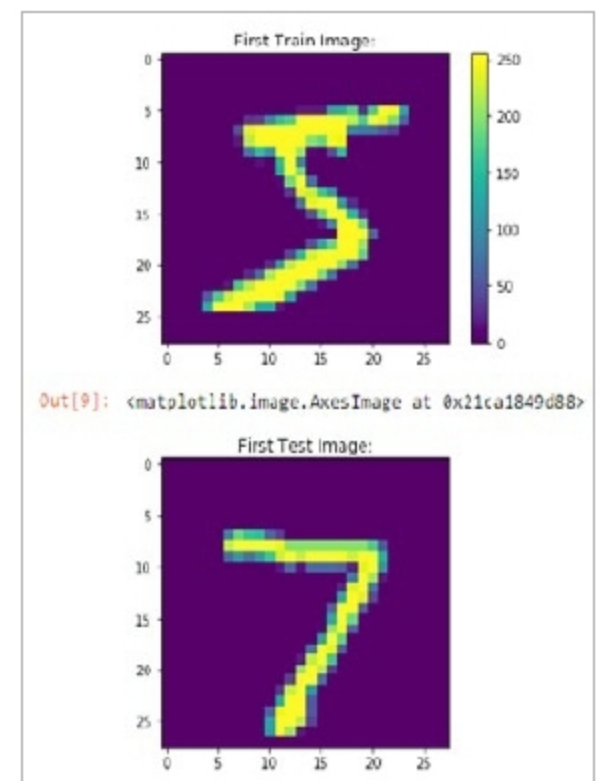


Figure 5: MNIST data set digit output

```
train my deep network, rest 10k will be
used for validation/testing
(train_images, train_labels), (test_
images, test_labels) = mnist.load_
data()
print (train_images.shape);
print (train_labels.shape);
print (train_labels);
```

```
# Lets see the images visually using
MatPlot
plt.figure()
plt.title('First Train Image:');
plt.imshow(train_images[0]);
plt.colorbar()
plt.grid(False)
plt.show();
plt.title('First Test Image:');
plt.imshow(test_images[0])
```

The output is as shown in Figure 5.

Building the neural network

We have chosen a simple sequential network with two dense layers.

```
## Building the TensorFlow Network
# As this is a categorical
classification (multi-class, single
label) problem, going with seq "Dense"
sort of deep networks
network = models.Sequential();
```